Smart Workout: A RAG-Assisted Decision Support and Business Intelligence System for Personalized Weight Training

Project Proposal — Business Intelligence and Analytics

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1. Background

Weight training is widely practiced for strength and fitness, yet many gym users struggle to plan effectively: they are uncertain which exercises suit a given body part, how to adapt to available equipment, how to structure a weekly timetable, and whether their physical condition supports a particular plan. Most decisions come from generic online routines that ignore personal factors — age, sleep, BMI, blood pressure, heart rate, equipment, schedule — producing a mismatch between plan and readiness. This project proposes **Smart Workout**, a combined Decision Support System (DSS) and Business Intelligence System (BIS) (Power, 2002; Chaudhuri et al., 2011) that unifies multi-source real-world data, a Retrieval-Augmented Generation (RAG) knowledge layer, and supervised ML models behind an interactive web dashboard, transforming raw input into descriptive insight, predictive outcome estimates, and prescriptive training plans while preserving human oversight. Intended end users are gym users seeking personalized training guidance; fitness coaches form a secondary audience using the dashboard for decision-support reference.

2. Objectives

The project will deliver a working DSS/BIS prototype that fulfils the following objectives, explicitly covering the descriptive, predictive, and prescriptive analytics layers required of a BI system:

- **Data collection & preparation:** clean and standardize multi-source public datasets (exercise catalogue, gym-member workout tracking, sleep & lifestyle, daily nutrition; details in §4) plus structured user input.
- **Descriptive analytics:** segment users into lifestyle profiles via K-Means clustering on the sleep dataset, and present multi-dataset exploratory insights (workout-type distribution, calorie-burn patterns, exercise coverage, nutrition macro mix) on the dashboard.
- **Predictive analytics:** train and compare three candidate models per task on the gym-tracking data, selecting the best by 5-fold cross-validation: (a) *calorie-burn-rate* regression (kcal / hr; session duration excluded so the model learns physiology); (b) *Workout Intensity* 3-class classification (Low / Mid / High, derived from the BPM-response ratio). A rule-derived *Training Readiness* band acts as an auxiliary feature (candidate models detailed in §5).
- **Prescriptive analytics:** generate a daily/weekly training plan via a rule-based engine conditioned on the predicted intensity, readiness, target body part, and available equipment; the predicted calorie-burn rate is displayed alongside the plan.
- **RAG knowledge retrieval & AI chat assistant:** retrieve exercise instructions, equipment-substitution rules, and timetable guidance; expose a pop-up AI chat widget that answers follow-ups and supports post-generation plan refinement (*e.g.* "Why this exercise?", "Substitute dumbbells", "Make Tuesday lighter").
- **BI dashboard:** React-based dashboard (with embedded Tableau views) presenting the multi-dataset descriptive view, personalized predictions, prescribed plan, RAG-sourced explanation, and embedded chat.

3. Description of the Proposed DSS/BIS

Smart Workout will be developed as a React + FastAPI web application integrating multi-source data, two best-performing predictive models (to be selected after comparing three candidates per task via 5-fold cross-validation; see §5), a ChromaDB-based RAG knowledge layer (Lewis et al., 2020), a rule-based plan generator, and an embedded RAG-grounded chat assistant (LLM-augmented when available). The selected regressor and classifier run on every user request and provide input to the prescriptive plan generator; the chat reuses the same RAG pipeline so its answers are grounded in the indexed corpus rather than parametric memory. Architecture and workflow appear in Figure 1 and the numbered steps below. Technology stack: React (frontend, with embedded Tableau dashboards), FastAPI (backend), pandas / scikit-learn / XGBoost (analytics), ChromaDB + `all-MiniLM-L6-v2` (RAG); a general-purpose LLM (*e.g.* OpenAI GPT, Grok, or a comparable provider) for the chat and explanation layer.

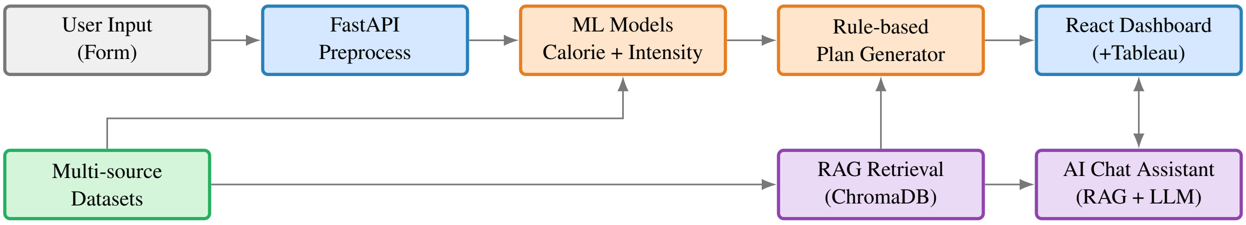

Figure 1: Smart Workout system architecture and workflow.

End-to-end workflow (Figure 1). The user enters input data via the React dashboard form → FastAPI preprocesses input, splits blood pressure into systolic and diastolic components, and derives the auxiliary readiness feature → the two selected models predict calorie-burn rate and intensity, while an exercise recommender filters ExerciseDB by body part and equipment → RAG retrieves top-k instructions and rule snippets → the rule-based generator assembles the plan and an optional LLM generates a short explanation → the dashboard renders the plan, predictions, and chat assistant.

4. Data Sources

Four real-world public datasets are integrated: an exercise catalogue (ExerciseDB, n.d.), gym workout sessions (Khorasani, 2024), sleep and lifestyle records (Tharmalingam, 2023), and daily nutrition (Shamim, 2025). Together they span structured tabular sessions, semi-structured exercise documents, and lifestyle records; the pipeline scales to wearable streams or gym-chain telemetry without redesign. Datasets are not joined by a shared key; each plays a distinct role.

Dataset	Records	Role in the system
ExerciseDB (raw flat) (ExerciseDB, n.d.)	1,500	Exercise catalogue: body parts, equipment, target muscles, step-by-step instructions. Used by the exercise recommender and embedded for RAG.
Gym Members Exercise Tracking (Khorasani, 2024)	973	Real workout sessions with calories burned, BPM (max/avg/resting), session duration, workout type, BMI, and experience level. Primary training data for the predictive models.
Sleep Health & Lifestyle (Tharmalingam, 2023)	374	Provides clusters for the descriptive lifestyle-profile view and supplies the rule-based *Training Readiness* auxiliary feature.
Daily Food Nutrition (Shamim, 2025)	651	Provides macro/calorie context shown on the dashboard and used in nutrition-aware rule snippets retrieved by RAG.

5. Methods: Descriptive, Predictive, Prescriptive

Descriptive. K-Means clustering on the Sleep & Lifestyle dataset identifies 3–4 user archetypes; the dashboard also presents cross-dataset exploratory views (workout-type distribution, calorie-burn patterns, exercise coverage, nutrition macro mix).

Predictive. For each predictive task, three candidate models well-suited to an approximately 1,000-row tabular problem are trained and compared, with the best selected by 5-fold cross-validation. The candidate set spans an interpretable linear baseline, a robust bagged-tree ensemble, and a state-of-the-art boosted-tree model:

- **Calorie-burn-rate regression** (kcal / hr; session duration excluded so the model learns physiology rather than the trivial duration–calories relationship, r=0.91): *Ridge Regression*, *Random Forest Regressor*, *XGBoost*

Regressor (Chen & Guestrin, 2016). Features: demographics, body composition, BPM, fat percentage, workout type, experience, water intake. Metrics: RMSE, MAE, R^2.

- **Workout-intensity classification** (3-class Low / Mid / High, derived from the average-to-maximum BPM ratio binned by tertiles ≤ 0.70, 0.70–0.85, ≥ 0.85): *Multinomial Logistic Regression*, *Random Forest Classifier*, *XGBoost Classifier*. Features: demographics, body composition, resting BPM, fat percentage, experience level, auxiliary readiness. Metrics: accuracy, macro-F1, confusion matrix.

SHAP (Lundberg & Lee, 2017) is applied to the selected models for feature-importance attribution shown in the dashboard.

Prescriptive. A rule-based generator combines predicted intensity, readiness, target body part, and the equipment-filtered exercise pool into a structured plan (split, exercises, sets, reps, rest). Intensity modulates volume and load (High $\rightarrow$ 4–5 sets, Low $\rightarrow$ deload/mobility focus). ChromaDB (indexing the 1,500 ExerciseDB instructions plus a curated rule corpus, embedded with `all-MiniLM-L6-v2`) supplies rationale and substitutions to both the plan generator and chat; the system retains full functionality without the LLM stage, ensuring graceful degradation. The LLM layer is provider-agnostic — any general-purpose AI model (*e.g.* OpenAI GPT, Grok) can be integrated.

6. Expected Outputs

A working React + FastAPI prototype with a ChromaDB vector store and the two best-performing trained models; a personalized training plan (JSON + dashboard view) including split, exercises, sets/reps, intensity band, predicted calorie-burn rate, and equipment substitutions; predictive insights with class probabilities and SHAP feature attribution; and a three-view BI dashboard (descriptive exploration, personalized prediction + plan, RAG-retrieved explanation) with an embedded RAG-grounded chat assistant.

7. Conclusion

Smart Workout applies DSS/BIS principles to personalized fitness planning by combining multi-source real-world data, supervised ML, RAG retrieval, and a rule-based prescriptive engine, producing a working prototype with evaluated, best-of-candidate models for the final report. It is a decision-support tool — not a clinical or coaching replacement; predictive outputs are reported with class probabilities, cross-validated error metrics, and transparent model limitations; readiness and intensity labels are rule-derived auxiliary constructs. The gym-tracking dataset is moderate, so cross-validated metrics will be reported transparently; recommendations are session-level. Delivery is staged: retrieval-only RAG, descriptive view, and plan view are the core deliverable, while LLM chat and SHAP visualization are graceful-degradation extensions.

References

Chaudhuri, S., Dayal, U., & Narasayya, V. (2011). An overview of business intelligence technology. *Communications of the ACM, 54*(8), 88–98. `https://doi.org/10.1145/1978542.1978562`

Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (pp. 785–794). ACM. `https://doi.org/10.1145/2939672.2939785`

ExerciseDB. (n.d.). *ExerciseDB: Free exercise database*. `https://exercisedb.dev/`

Khorasani, V. (2024). *Gym members exercise dataset* [Data set]. Kaggle. `https://www.kaggle.com/datasets/valakhorasani/gym-members-exercise-dataset`

Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W., Rocktäschel, T., Riedel, S., & Kiela, D. (2020). Retrieval-augmented generation for knowledge-intensive NLP tasks. *Advances in Neural Information Processing Systems, 33*, 9459–9474.

Lundberg, S. M., & Lee, S.-I. (2017). A unified approach to interpreting model predictions. *Advances in Neural Information Processing Systems, 30*, 4765–4774.

Power, D. J. (2002). *Decision support systems: Concepts and resources for managers*. Quorum Books.

Shamim, A. (2025). *Daily food and nutrition dataset* [Data set]. Kaggle. `https://www.kaggle.com/datasets/adilshamim8/daily-food-and-nutrition-dataset`

Tharmalingam, L. (2023). *Sleep health and lifestyle dataset* [Data set]. Kaggle. `https://www.kaggle.com/datasets/uom190346a/sleep-health-and-lifestyle-dataset`